

Wenyu Du

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EDUCATION

- **University of Hong Kong** Mar. 2021 – Now
Doctor of Philosophy in Computer Science
- **University of Southampton** Sep. 2016 – Nov. 2017
Master of Science in Data Science with Distinction
- **University of Nottingham** Sep. 2011 – July. 2016 (Sep. 2013 – Sep. 2014, Sick leave)
Bachelor of Science in Computer Science with High 2:1

SELECTED PUBLICATIONS

- Stephen Chung*, **Wenyu Du***, Jie Fu. “*Thinker: Learning to Think Fast and Slow*”, Preprint, 2025.
- Yifan Zhang, **Wenyu Du**, Dongming Jin, Jie Fu, Zhi Jin. “*Finite State Automata Inside Transformers with Chain-of-Thought: A Mechanistic Study on State Tracking*”, (ACL, Oral), 2025.
- **Wenyu Du***, Tongxu Luo*, Zihan Qiu, Zeyu Huang, Yikang Shen, Reynold Cheng, Yike Guo, Jie Fu, “*Stacking Your Transformers: A Closer Look at Model Growth for Efficient LLM Pre-Training*”, Accepted to The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS, Spotlight), 2024.
- **Wenyu Du***, Shuang Cheng*, Tongxu Luo, Zihan Qiu, Zeyu Huang, Ka Chun Cheung, Reynold Cheng, Jie Fu, “*Unlocking Continual Learning Abilities in Language Models*”, Accepted to The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP, Findings), 2024.
- Jacob Louis Hoover, **Wenyu Du**, Alessandro Sordoni, Timothy J. O’Donnell, “*Linguistic Dependencies and Statistical Dependence*”, The 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP, Main), 2021.
- **Wenyu Du***, Zhouhan Lin*, Yikang Shen, Timothy J. O’Donnell, Yoshua Bengio, Yue Zhang, “*Exploiting Syntactic Structure for Better Language Modeling: A Syntactic Distance Approach*”, The 58th Annual Meeting of the Association for Computational Linguistics (ACL, Main), 2020.

* = equal contribution.

PUBLICATIONS

- **Wenyu Du**, Rui Zheng, Tongxu Luo, Stephen Chung, Jie Fu “*Fine-Grained Manipulation of Arithmetic Neurons*”, Preprint, 2025.
- Stephen Chung*, **Wenyu Du***, Jie Fu. “*Thinker: Learning to Think Fast and Slow*”, Preprint, 2025.
- Tongxu Luo*, **Wenyu Du***, Jiayi Bi, Stephen Chung, Zhengyang Tang, Hao Yang, Min Zhang, Benyou Wang. “*Learning from Peers in Reasoning Models*”, Preprint, 2025.
- Stephen Chung*, **Wenyu Du***, Jie Fu. “*Learning from Failures in Multi-Attempt Reinforcement Learning*”, Preprint, 2025.
- Tongxu Luo*, **Wenyu Du***, Hanwen Hao, Min Zhang, Hao Yang, Benyou Wang. “*E2-RAG: Towards Editable Efficient RAG by Editing Compressed KV Caches*”, COLM, 2025.

- Xu Wang, Yan Hu, **Wenyu Du**, Reynold Cheng, Benyou Wang, Difan Zou. “*Towards Understanding Fine-Tuning Mechanisms of LLMs via Circuit Analysis*”, Accepted to The Forty-Second International Conference on Machine Learning (ICML, Poster), 2025.
- Yifan Zhang, **Wenyu Du**, Dongming Jin, Jie Fu, Zhi Jin. “*Finite State Automata Inside Transformers with Chain-of-Thought: A Mechanistic Study on State Tracking*”, (ACL, Oral), 2025.
- **Wenyu Du**^{*}, Tongxu Luo^{*}, Zihan Qiu, Zeyu Huang, Yikang Shen, Reynold Cheng, Yike Guo, Jie Fu, “*Stacking Your Transformers: A Closer Look at Model Growth for Efficient LLM Pre-Training*”, Accepted to The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS, Spotlight), 2024.
- **Wenyu Du**^{*}, Shuang Cheng^{*}, Tongxu Luo, Zihan Qiu, Zeyu Huang, Ka Chun Cheung, Reynold Cheng, Jie Fu, “*Unlocking Continual Learning Abilities in Language Models*”, Accepted to The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP, Findings), 2024.
- Ka Man Lo, Yiming Liang, **Wenyu Du**, Yuantao Fan, Zili Wang, Wenhao Huang, Lei Ma, Jie Fu, “*m2mKD: Module-to-Module Knowledge Distillation for Modular Transformers*”, arXiv:2402.16918.
- Jinyang Li, Binyuan Hui, Reynold Cheng, Bowen Qin, Chenhao Ma, Nan Huo, Fei Huang, **Wenyu Du**, Luo Si, Yongbin Li, “*Graphix-T5: Mixing Pre-Trained Transformers with Graph-Aware Layers for Text-to-SQL Parsing*”, Accepted to The 37th AAAI Conference on Artificial Intelligence (AAAI, Oral) 2023.
- Yuqiao Wen, Zichao Li, **Wenyu Du**, Lili Mou, “*f-Divergence Minimization for Sequence-Level Knowledge Distillation*”, The 61st Annual Meeting of the Association for Computational Linguistics (ACL, Main), 2023.
- Jacob Louis Hoover, **Wenyu Du**, Alessandro Sordoni, Timothy J. O’Donnell, “*Linguistic Dependencies and Statistical Dependence*”, The 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP, Main), 2021.
- Qiankun Fu, Linfeng Song, **Wenyu Du**, Yue Zhang, “*End-to-End AMR Coreference Resolution*”, The 59th Annual Meeting of the Association for Computational Linguistics (ACL, Main), 2021.
- **Wenyu Du**^{*}, Zhouhan Lin^{*}, Yikang Shen, Timothy J. O’Donnell, Yoshua Bengio, Yue Zhang, “*Exploiting Syntactic Structure for Better Language Modeling: A Syntactic Distance Approach*”, The 58th Annual Meeting of the Association for Computational Linguistics (ACL, Main), 2020.
- Sushen Zhang, Ruijuan Chen, **Wenyu Du**, Ye Yuan, Vassilios S. Vassiliadis, “*A Hessian-Free Gradient Flow (HFGF) Method for the Optimisation of Deep Learning Neural Networks*”, Computers and Chemical Engineering (Journal), 2020.
- **Wenyu Du**, Baocheng Li, Min Yang, Qiang Qu, “*A Multi-Task Learning Approach for Answer Selection: A Study and a Chinese Law Dataset*”, The 33rd AAAI Conference on Artificial Intelligence (AAAI, student abstract), 2019.
- **Wenyu Du**, Shuai Yu, Min Yang, Qiang Qu, Jia Zhu, “*GPSP: Graph Partition and Space Projection based Approach for Heterogeneous Network Embedding*”, The 27th International World Wide Web Conference (WWW, poster), 2018.

^{*} = equal contribution.

SELECTED RESEARCH EXPERIENCE

- **NVIDIA** Jun. 2024 - Aug. 2024
 - *Research Intern*
 - Project: Unlocking continual learning ability in language models, accepted by EMNLP 2024
- **Westlake University** Sep. 2018 - Feb. 2021
 - *Research Assistant*
 - Project: Exploiting Syntactic Structure for Better Language Modeling: A Syntactic Distance Approach, accepted by ACL 2020

PROGRAMMING LANGUAGES

- **Deep Learning:** Tensorflow, Pytorch
- **Languages/Skills:** Python, Java, MATLAB, C, C++

LANGUAGE CERTIFICATE

- **IELTS:** Overall: 7.5 (Listening: 8.0, Speaking: 6.5, Reading: 8.5, Writing: 6.5)